

TPC-411: Pooled Odd Complete-Shells

Liang Wang

School of Mathematics and Statistics, Huazhong University
of Science and Technology (HUST), Wuhan, China

September 6, 2026

Abstract

TPC-409 and TPC-410 audited odd complete shells separately. We now pool the two full shells at $Q = 65536$ and $Q = 131072$, retaining all $5709 + 10749 = 16458$ primes. At fixed $H = 66$, $N = 264 = 4H$, the pooled profile has equal parity counts $m_- = m_+ = 8229$. Shell-local amplitude normalizations are kept, and the exact local identity plus Cauchy-Schwarz gives $0 \leq z \leq t_1/(a_{\min}\sqrt{S_0S_1}) \leq 4/(a_{\min}H) \leq 4/H$. An exact rational certificate and an independent literal CRT replay verify the pooled row. This is a synthetic finite proxy result, not an arithmetic or twin-prime theorem.

1 Pooled complete-shell profile

Let $\mathcal{P}$ be the increasing concatenation of the complete shells $(65536, 131072]$ and $(131072, 262144]$. Their counts are 5709 and 10749, so $|\mathcal{P}| = 16458$. For $p_i \in \mathcal{P}$, let Q_i denote its shell scale and set

$$H = 66, \quad N = 264 = 4H, \quad a_i = \frac{p_i^3}{Q_i^2(p_i - 1)}.$$

Choose a CRT representative $o > 10^6$ satisfying

$$o \equiv 0 \pmod{p_i} \quad (i \text{ even}), \quad o \equiv -N \pmod{p_i} \quad (i \text{ odd}).$$

All primes are retained; hence $m_- = m_+ = 8229$. With

$$t_d = \frac{H^2}{H^2 + d^2}, \quad S_0 = \sum_{d=1}^{N-1} t_d^2, \quad S_1 = \sum_{d=1}^{N-2} t_d^2 + t_1^2,$$

the local proxy has

$$G_0 = V_- S_0, \quad G_1 = V_- S_1 + V_+(S_1 - t_1^2), \quad M = t_1 P_-.$$

The parameter H is a proxy height, not the physical h_0 .

2 Finite pooled theorem

Theorem. If $a_{\min} = \min_i a_i$ and $z = M/\sqrt{G_0G_1}$, then

$$0 \leq z \leq \frac{t_1}{a_{\min}\sqrt{S_0S_1}} \leq \frac{4}{a_{\min}H} \leq \frac{4}{H}. \quad (1)$$

Proof. Every pooled prime is larger than N . The even class masks offset zero and the odd class first hits offset N , so no prime masks offset one and the three displayed identities follow. Since $m_- = 8229$, we have $P_-^2 \leq m_- V_-$ and $V_- \geq m_- a_{\min}^2$; moreover $G_1 \geq V_- S_1$. Thus

$$z^2 \leq \frac{t_1^2 m_-}{V_- S_0 S_1} \leq \frac{t_1^2}{a_{\min}^2 S_0 S_1}.$$

The terms $1 \leq d \leq H$ occur in both sums and satisfy $t_d \geq 1/2$, hence $S_0, S_1 \geq H/4$. Finally $a_i = (p_i/Q_i)^2 p_i/(p_i - 1) > 1$. Taking square roots proves (1). $\square$

3 Exact certificate and observation

The producer enumerates both full shells, records each prime's Q_i , and uses integer CRT and exact rational arithmetic. The pooled row has

shells	$ \mathcal{P} $	m_-	m_+	z
65536, 131072	16458	8229	8229	0.005216872683870

and the finite observation $H z = 0.344313597135425$. This decimal is derived from an exact rational square and is not an asymptotic claim. The independent checker rebuilds both sieves and the pooled CRT, then literally visits every per-prime, per-coordinate mask for both local rows before comparing G_0 , G_1 , and M .

4 Route boundary

This is one finite pooled synthetic adjacent normalized proxy entry. It does not prove a full normalized operator estimate, identify physical h_0 or arithmetic signs, pay arithmetic L^2 or fixed-power credit, close Route A/B, or imply a twin-prime result. The exact finite pooled odd complete-shell status is recorded in the certificate; arithmetic advance is NO, fixed-power credit is 0, and full Gate B is OPEN.

Reproduction

Run the producer, independent replay, and nine-mutation stress checker with `--check` in normal and optimized modes. Bridge-B repeats these checks, requires empty stderr and identical output, and locks the release artifacts by SHA-256.